

EE-2003 Computer Organization and Assembly Language

Monday, April 10, 2023

Course Instructor

Mr. Ali Raza

Serial No:

2nd Sessional Exam

Total Time: 1 Hour

Total Marks: 40

Signature of Invigilator

SOLUTION_____

Roll No

Section

Signature

DO NOT OPEN THE QUESTION BOOK OR START UNTIL INSTRUCTED.

Instructions:

1. Verify at the start of the exam that you have a total of **four (4)** questions printed on **Seven (7)** pages including this title page.
2. Attempt all questions on the question-book.
3. The exam is closed books, closed notes. Please see that the area in your threshold is free of any material classified as 'useful in the paper' or else there may a charge of cheating.
4. Write Anything you learned apart from COAL to get one bonus mark.
5. Read the questions carefully for clarity of context and understanding of meaning and make assumptions wherever required, for neither the invigilator will address your queries, nor the teacher/examiner will come to the examination hall for any assistance.
6. Fit in all your answers in the provided space. You may use extra space on the last page if required. If you do so, clearly mark question/part number on that page to avoid confusion.
7. Use only your own stationery and calculator. If you do not have your own calculator, use manual calculations.
8. Use only permanent ink-pens. Only the questions attempted with permanent ink-pens will be considered. Any part of paper done in lead pencil cannot be claimed for checking/rechecking.

	Q-1	Q-2	Q-3	Q-4	Total
Total Marks	10	10	9	11	40
Marks Obtained					

National University of Computer and Emerging Sciences

Department of **Computer Science**

Chiniot-Faisalabad Campus

Vetted By: _____ Vetter Signature: _____

Question	Answer
1	D
2	C
3	B
4	C
5	E
6	D
7	A
8	B
9	D
10	A

Note: Cutting and Over righting will results in zero marks.

Question 1 [10 Marks]

MCQs

- 1- Overflows occur when
 - a. Adding two positive numbers and the sum is negative
 - b. Adding two negative numbers and the sum is positive
 - c. Adding or subtracting unsigned integers
 - d. Both A and B
 - e. Both B and C
- 2- The maximum signed integers which can be stored on a 8 bit accumulator is
 - a. -128
 - b. 0
 - c. 127
 - d. 128
 - e. 255
- 3- _____ interrupt is used for DOS services
 - a. 20h
 - b. 21h
 - c. 16h
 - d. 18h
 - e. 01h
- 4- In computers, subtraction is carried out generally by_____.
 - a. 1's complement method
 - b. signed magnitude method
 - c. 2's complement method
 - d. BCD subtraction method
 - e. registers
- 5- When a subroutine is called, the address of the instruction following the CALL instructions stored in/on the _____.
 - a. return pointer
 - b. stack pointer
 - c. accumulator
 - d. program counter
 - e. stack
- 6- When the RET instruction at the end of subroutine is executed,
 - a. the information where the stack is initialized is transferred to the stack pointer
 - b. the memory address of the RET instruction is transferred to the program counter
 - c. two data bytes stored in the top two locations of the stack are transferred to the stack pointer
 - d. two data bytes stored in the top two locations of the stack are transferred to the program

counter

e. All of the above

- 7- OFFSET returns the distance in bytes, of a label from the beginning of its enclosing segment
- a. True
 - b. False
- 8- The USES operator only generates PUSH instructions, so you must code POP instructions yourself.
- a. True
 - b. False
- 9- What is the value in bl after performing the following operations?
- ```
mov bl, 88h
rcr bl, 3
```
- a. 10001000b
  - b. 01000100b
  - c. 00100001b
  - d. 00010001b
  - e. 00010010b
- 10- What is the value of ax after performing the following operations?
- ```
mov ax, 0EE21h  
shr ah,02
```
- a. 01D21h
 - b. 001Dh
 - c. 0DC21h
 - d. 00DCh
 - e. 0211Dh

Question 2 [10 Marks]

Write a procedure **ARRAY_SUM** that sums the **ARRAY** given below. Parameters are passed through registers and so is the return value. Parameters include **ARRAY**, **ARRAY SIZE** and the total is saved in **SUM**.

ARRAY dw 5h,10H,6,7,01010b

SUM db 0

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Question 3 [6 + 3 = 9 Marks]

Part A [3+3 = 6 Marks]

What does the following code do? Be clear in your answer and give an example with ***BX*** = ***734Bh***. Provide values of BX and DX at the end of program.

```
mov dx, 0
mov cx, 16
L1:
shl bx, 1
rcr dx, 1
loop L1
```

- 1 – Reversing the bits of DX. Putting 0 value in this case
- 2- BX will be zero, due to logical shifts of 16 bits.

Part B [3 Marks]

Implement the following C++ expression in assembly language, using 16-bit unsigned operands:

```
val1 = (val2 * val3) / (val4 - 3)
```

Note: Right only .code segment , assuming variables are already declared and initialized in .data segment.

```
Mov ax, val2
Mul val3

Sub val4,3
Div val4

Mov val1,AX
```

Question 4 [2+2+3+2+2 = 11 Marks]

- 1- Write a single rotate instruction that exchanges the high and low halves of the DL register.

```
ROR DL, 4 ; rotate the contents of DL to the right by 4 bits
```

- 2- Write a logical shift instruction that divides EBX by 4

```
SHR EBX,2
```

- 3- Write a sequence of shift instructions that cause AX to be sign-extended into EAX. In other words, the sign bit of AX is copied into the upper 16 bits of EAX

```
SAL AX, 16 ; shift AX left by 16 bits  
SAR EAX, 16 ; shift EAX right by 16 bits, preserving the sign bit
```

- 4- What will be the contents of AX and DX after the following operation?

```
mov ax, 4000h  
mov dx, 500h  
mov bx, 10h  
div bx
```

```
AX : 400 h    ; Qoutient  
DX : 0h       ; Remainder
```

- 5- What will be the contents of AX and DX after the following operation?

```
mov dx, 0  
mov ax, 222h  
mov cx, 100h  
mul cx
```

```
AX : 22000h  
DX : 0
```